Speech Analysis

Waveform & Amplitude

Waveform

Sound signal over time

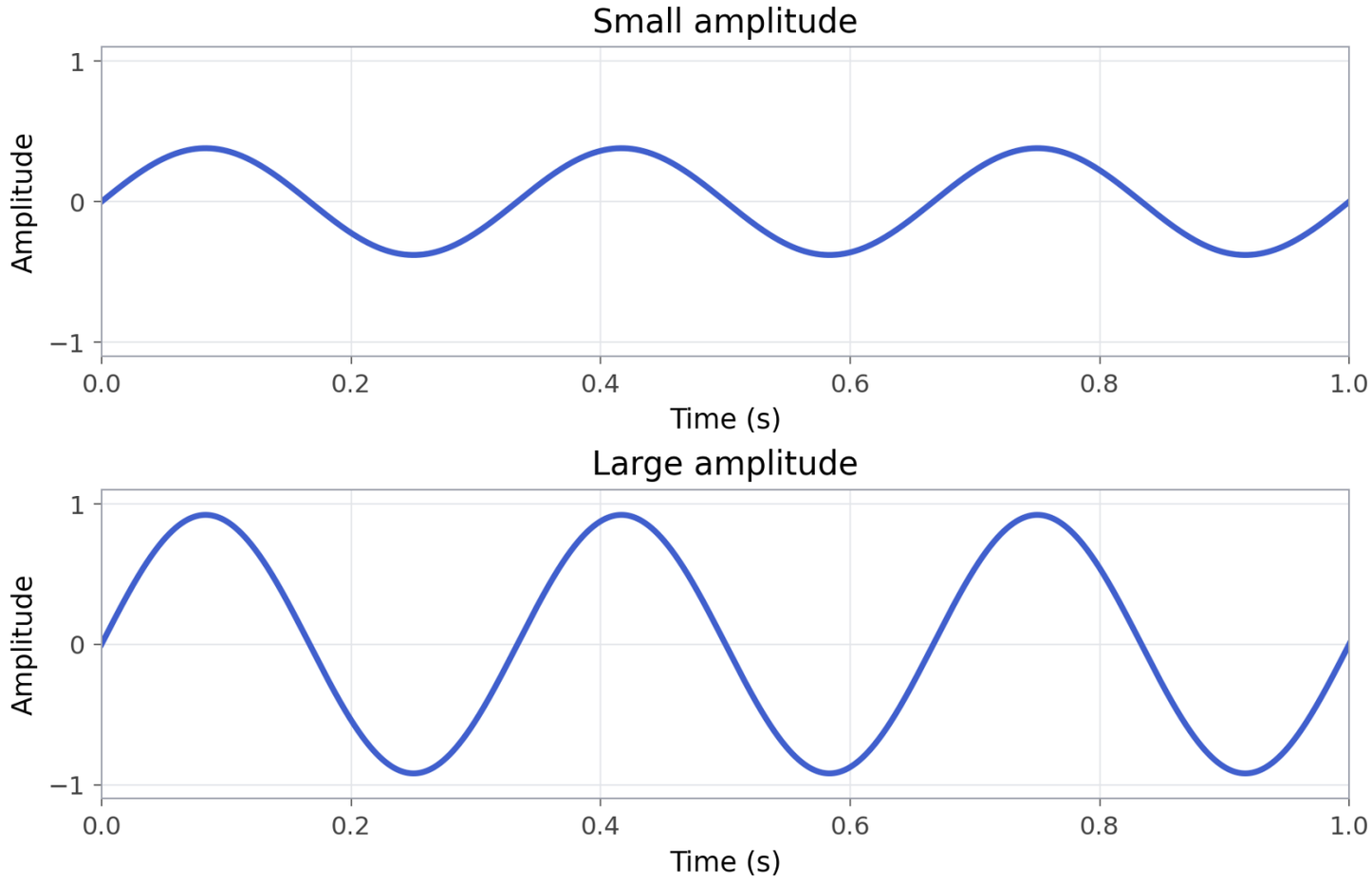
X-axis

Time (seconds)

Y-axis

Amplitude (signal size)

Larger amplitude usually means a louder sound.



Frequency

Frequency

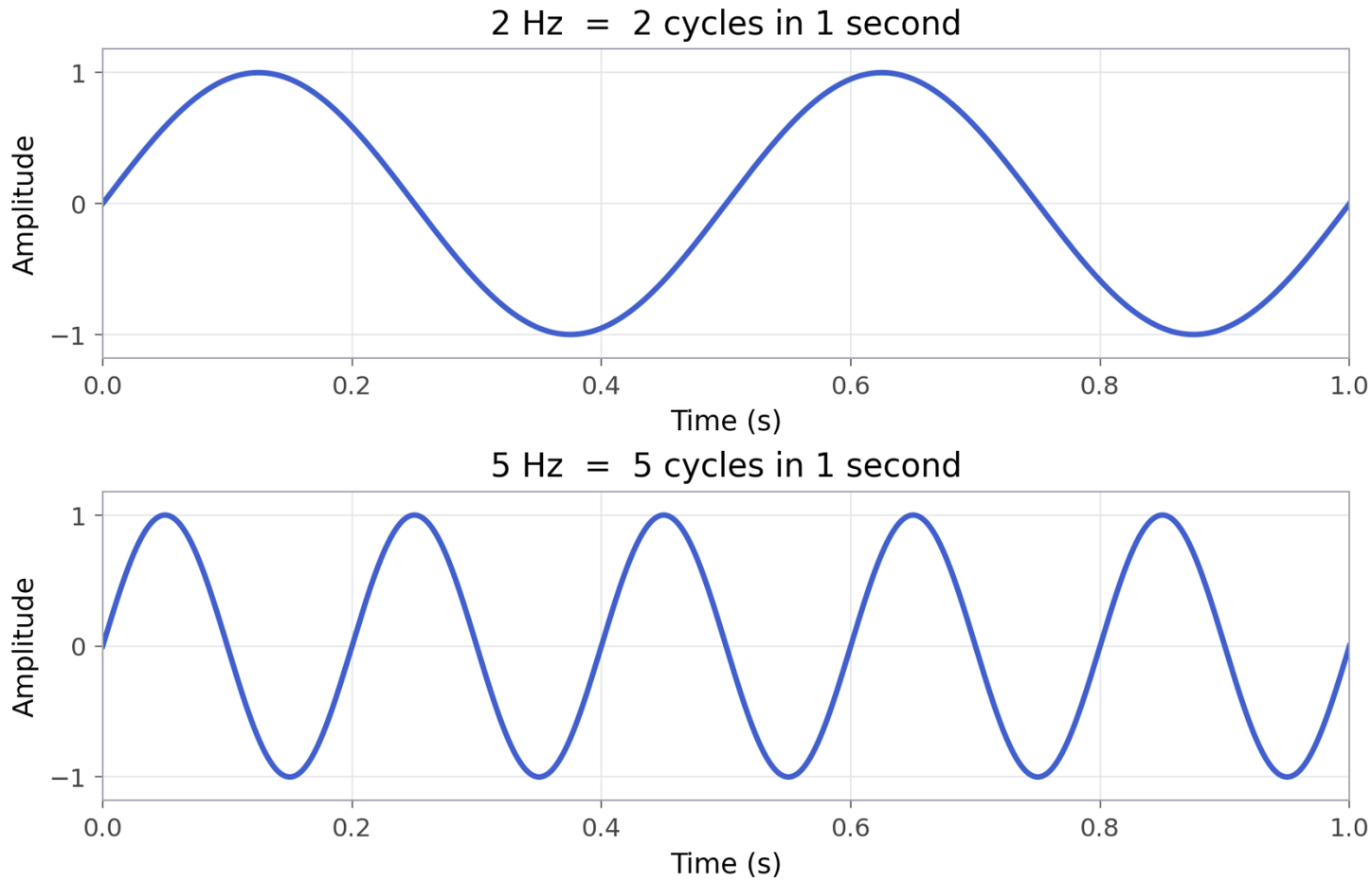
Cycles per second

Unit

Hertz (Hz)

Higher frequency

Higher pitch



Sampling Rate

Sampling rate

Measurements per second

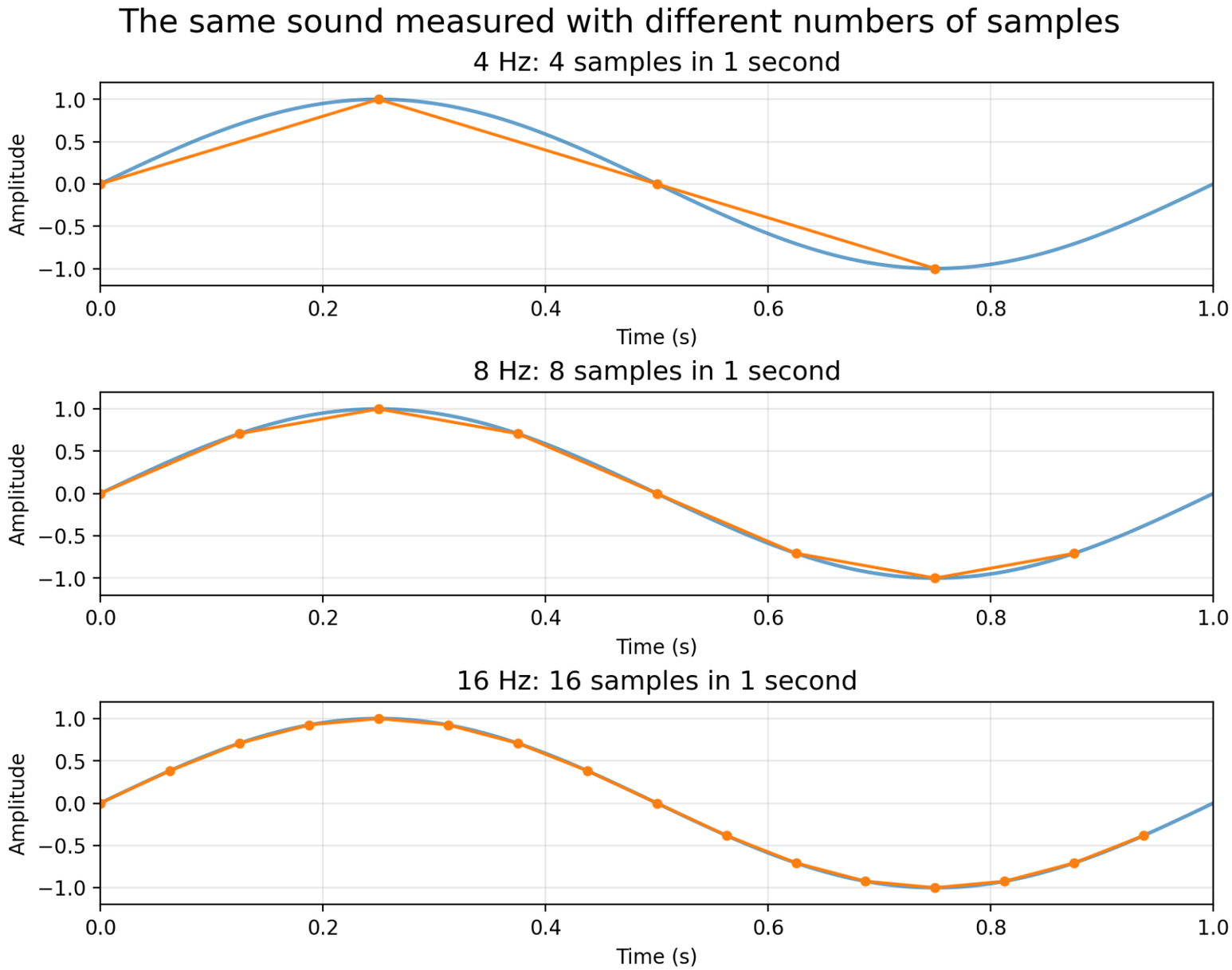
Unit

Hertz (Hz)

More samples

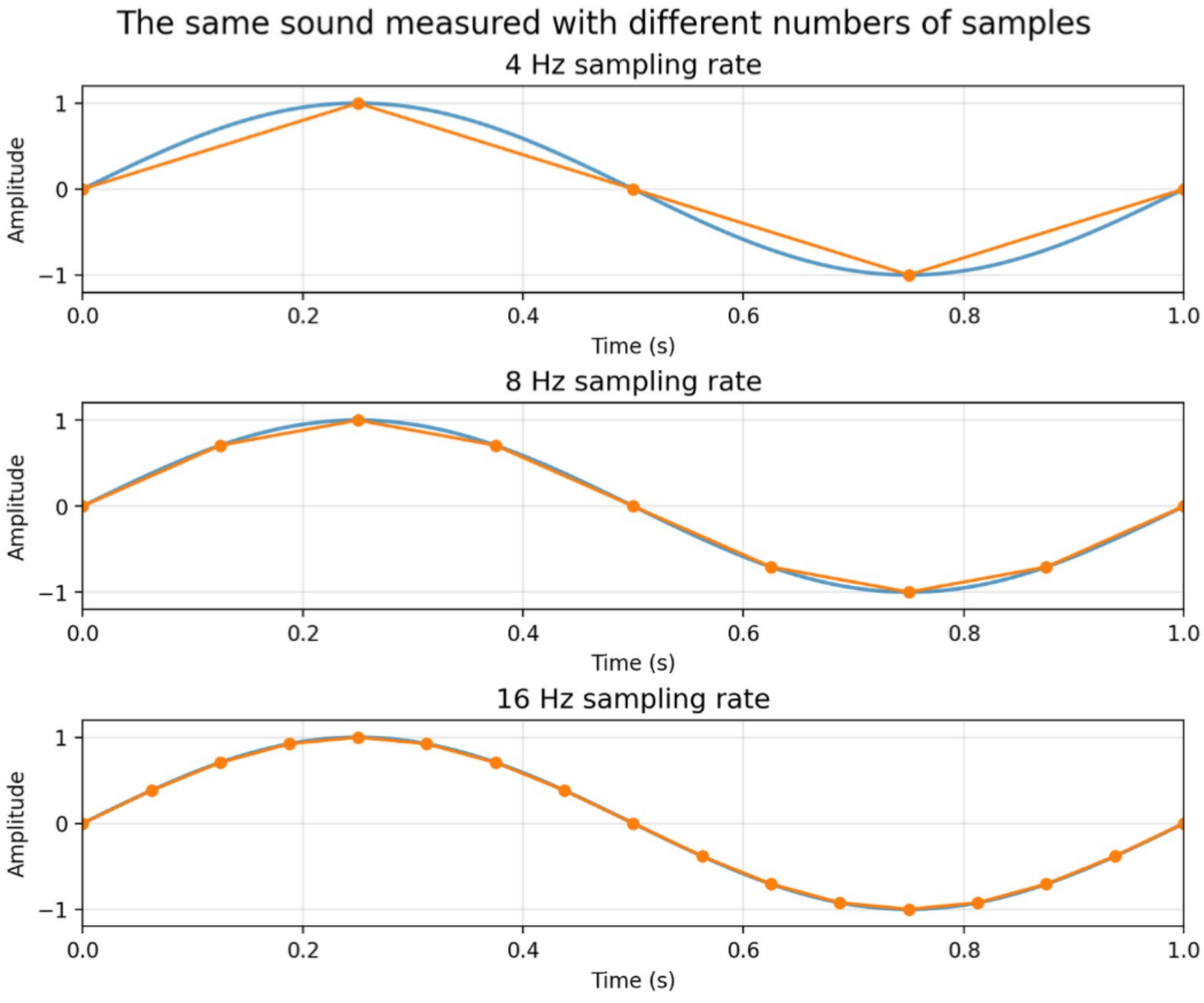
More points on the same wave

The orange dots are the sampled values.



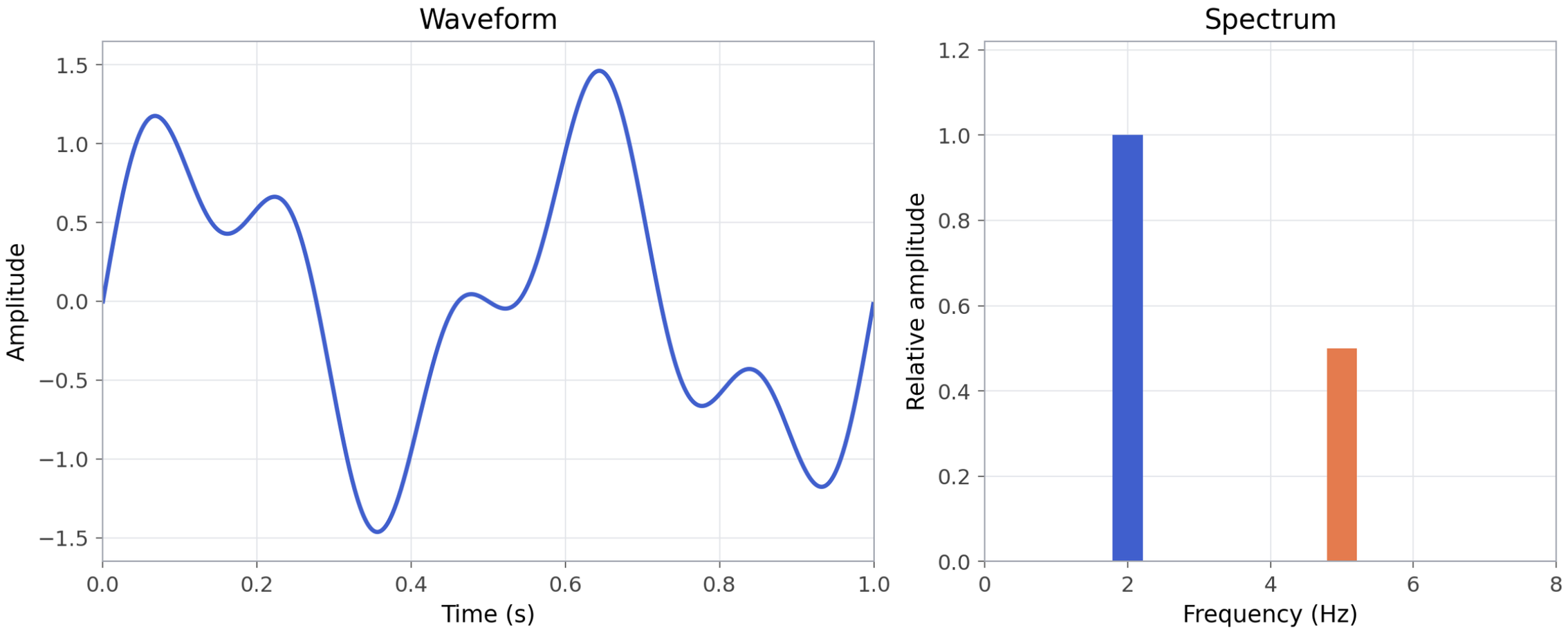
Frequency describes the sound; sampling rate describes digital recording.

Sampling Rate



Spectrum

One sound can contain several frequency components.



Waveform

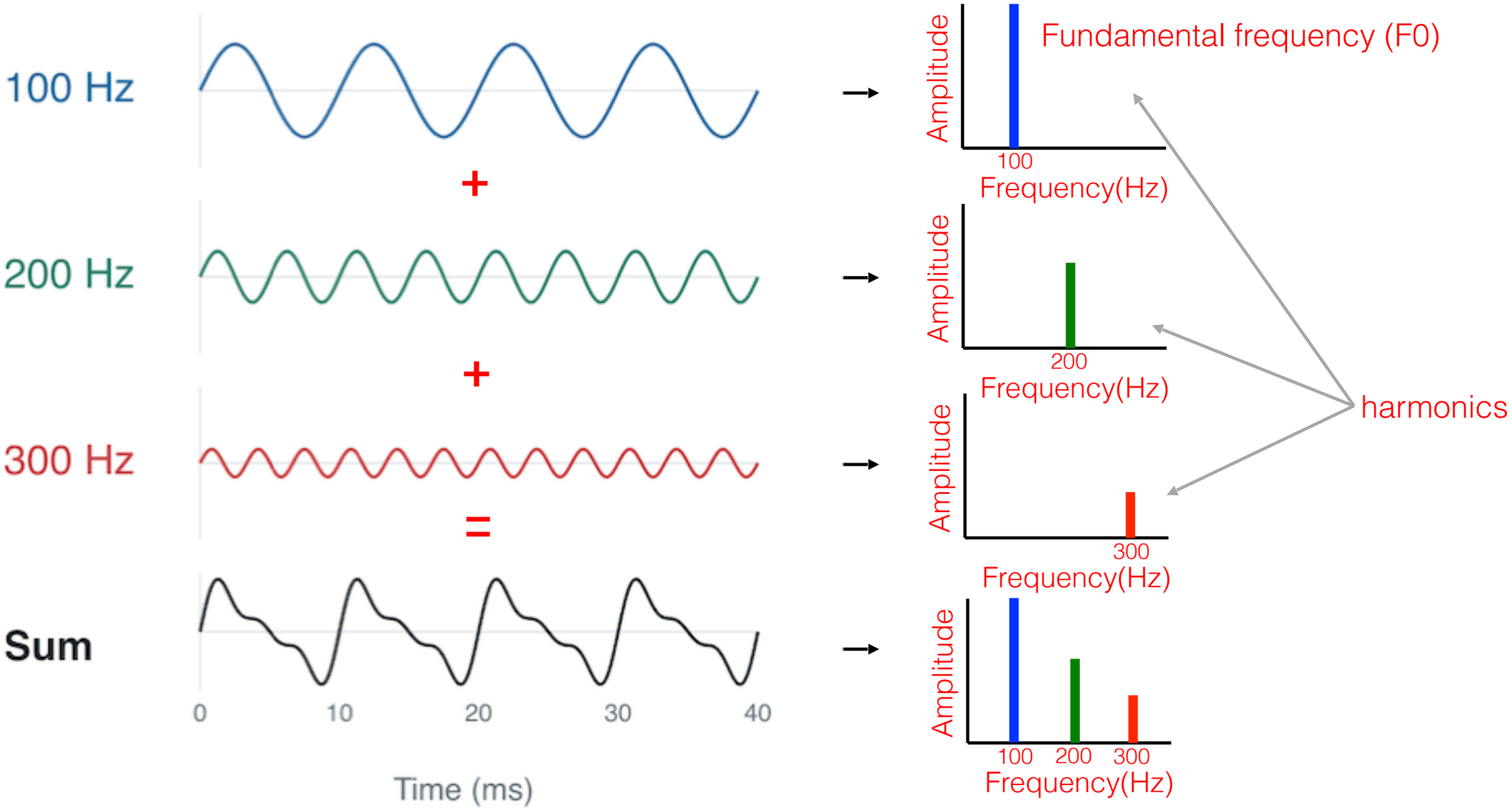
How the signal changes over time

Spectrum

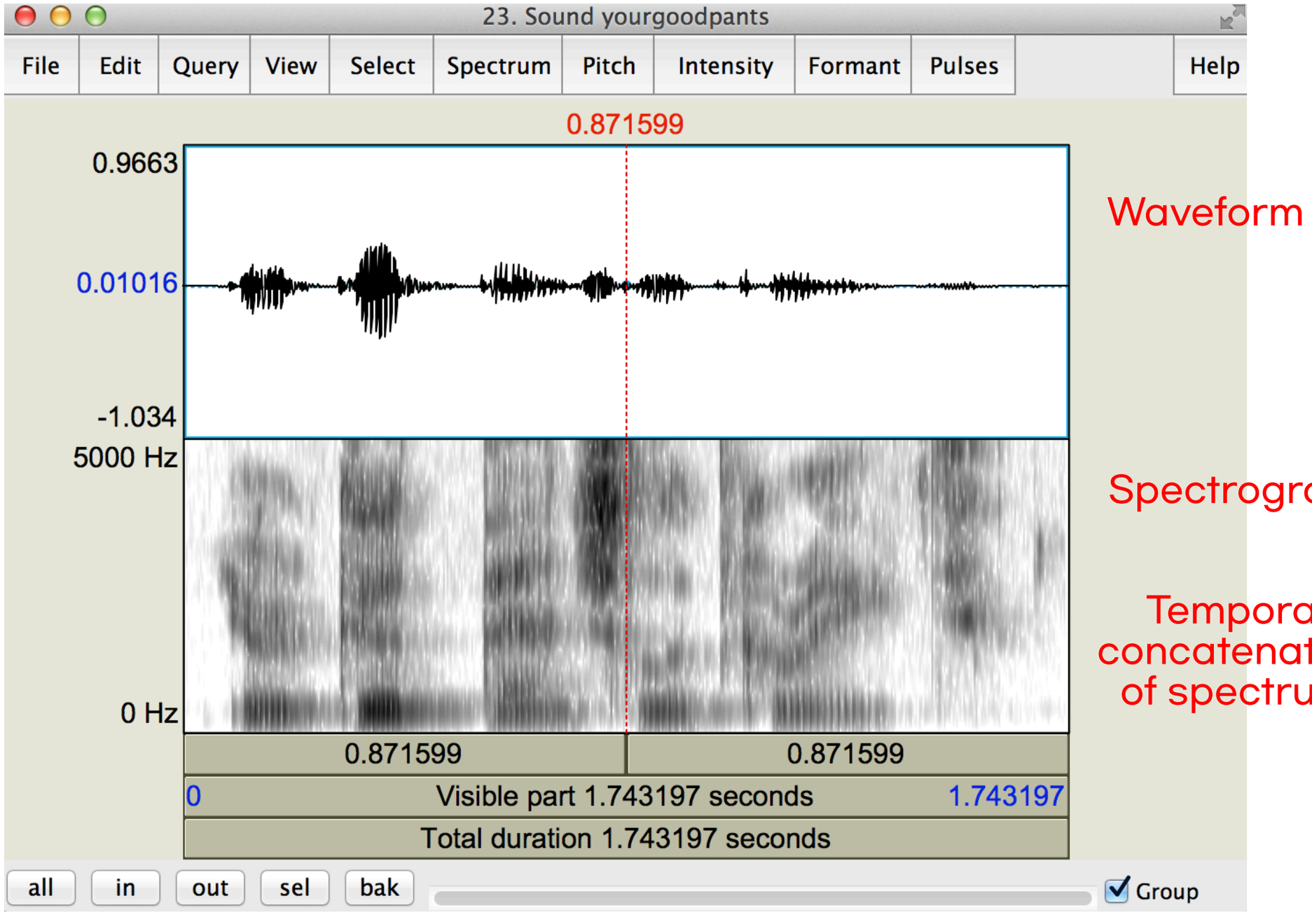
How strong each frequency is

Illustration: a 2 Hz component plus a weaker 5 Hz component.

Spectrum


100 Hz
+
200 Hz
+
300 Hz
=
Sum
0
10
20
30
40
Time (ms)
Amplitude
Fundamental frequency (F0)
100
Frequency(Hz)
Amplitude
200
Frequency(Hz)
harmonics
Amplitude
300
Frequency(Hz)
Amplitude
100
200
300
Frequency(Hz)

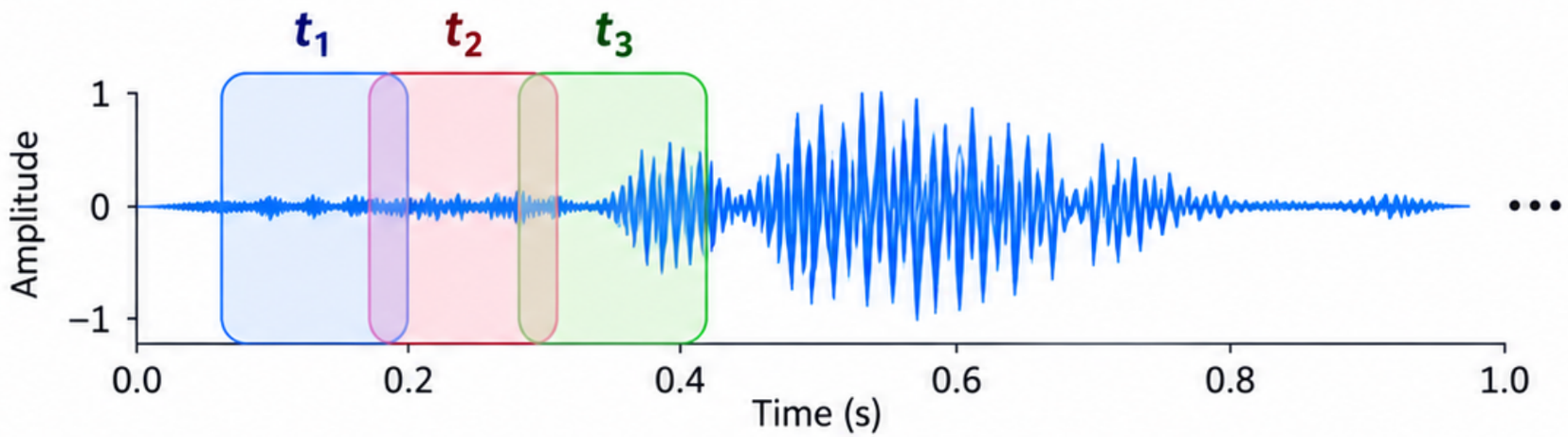
Spectrogram



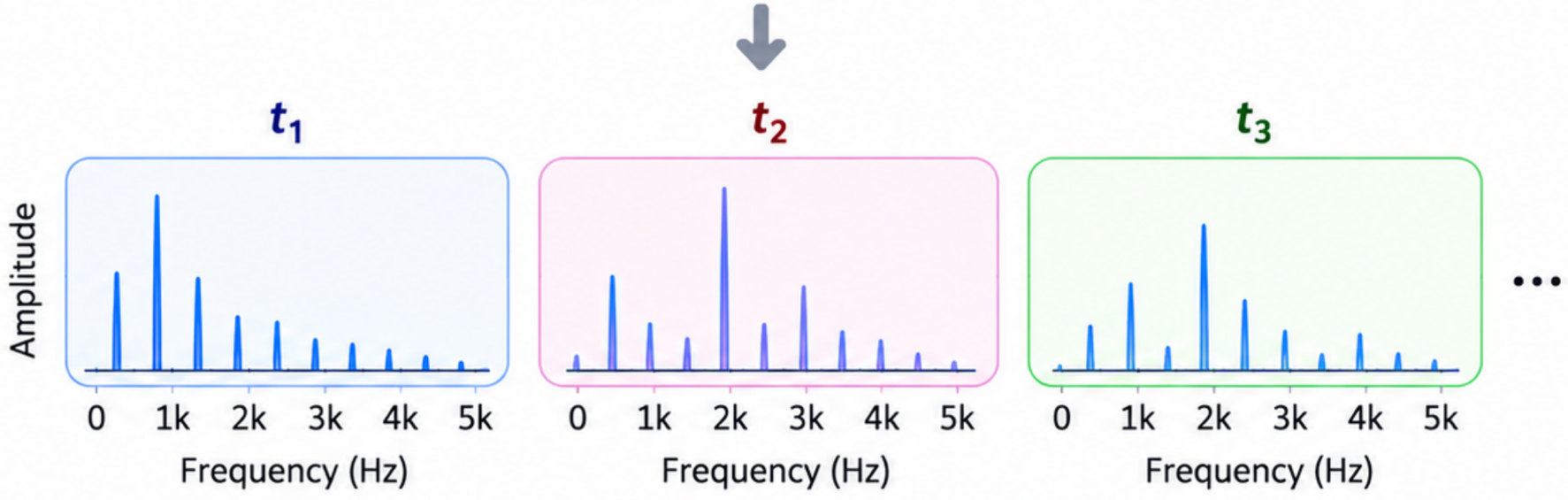
Spectrogram

A spectrogram is made by computing a spectrum for short, overlapping time windows and stacking them in time.

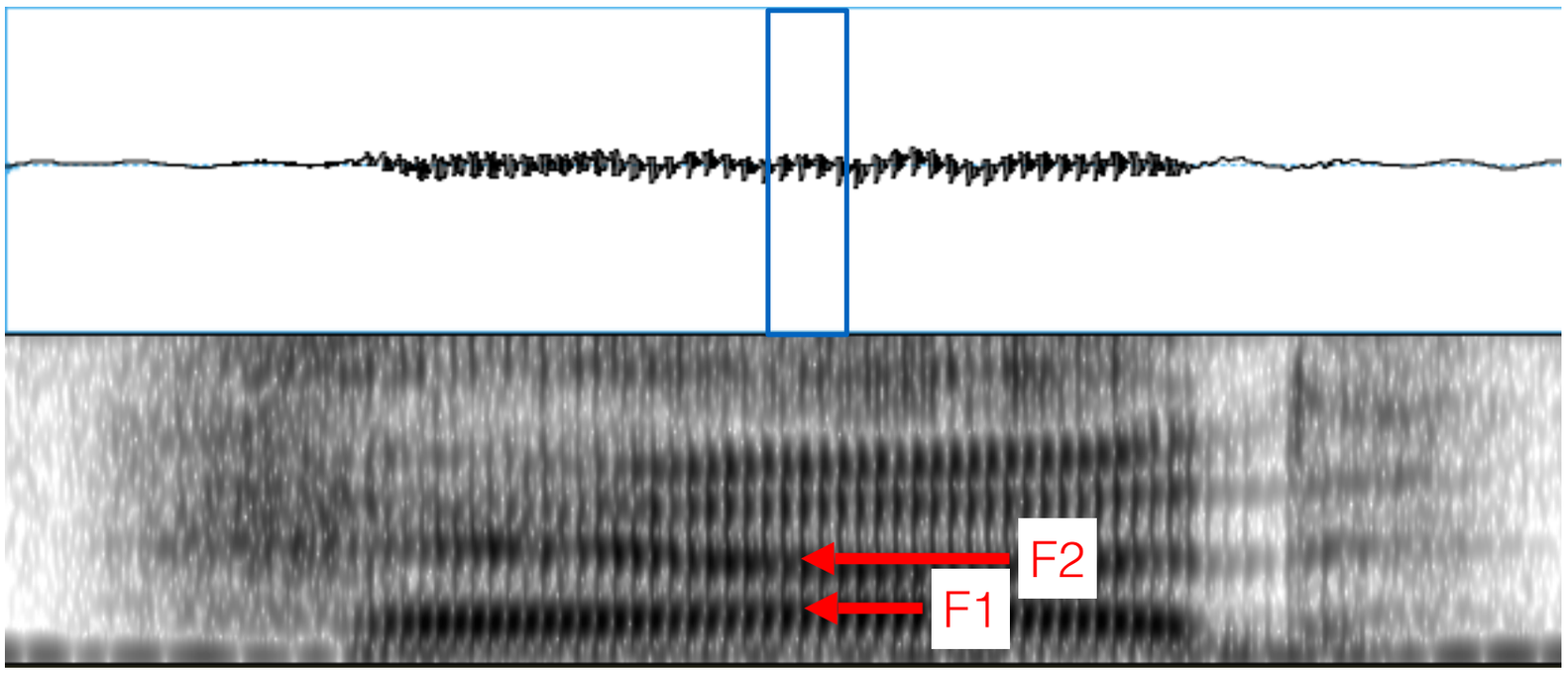
1 **Split the waveform into short windows**
(e.g., 20–30 ms)

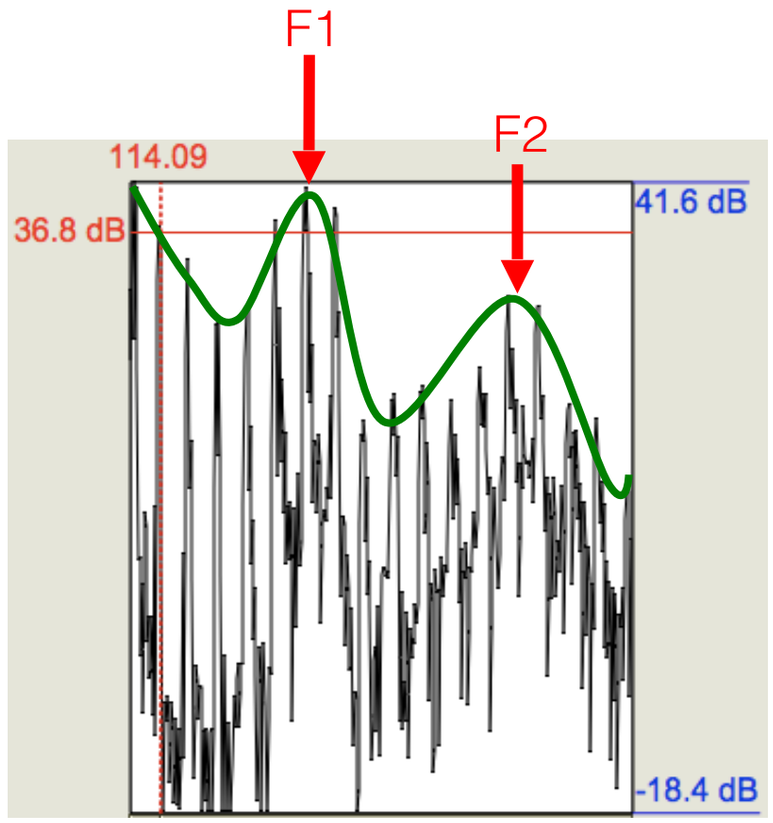


2 **Compute a spectrum for each window**



Spectrogram





Airplane view of temporal concatenation of spectrum

Dark band: mountains = Formants

Three Ways to See a Sound

Waveform

X = Time

Y = Amplitude

Changes in signal size

Spectrum

X = Frequency

Y = Amplitude

Frequency components

Spectrogram

X = Time

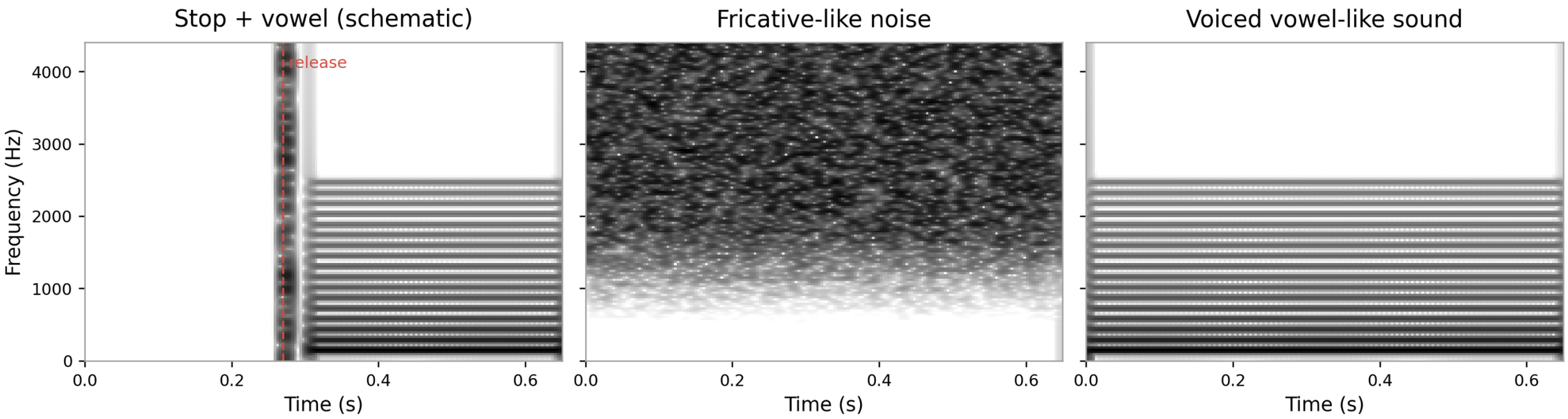
Y = Frequency

Energy shown by darkness

A spectrogram combines time and frequency information.

Reading Speech Spectrograms

Look for changes in time and frequency, not just the overall shape.



Stop

Closure + brief release

Fricative

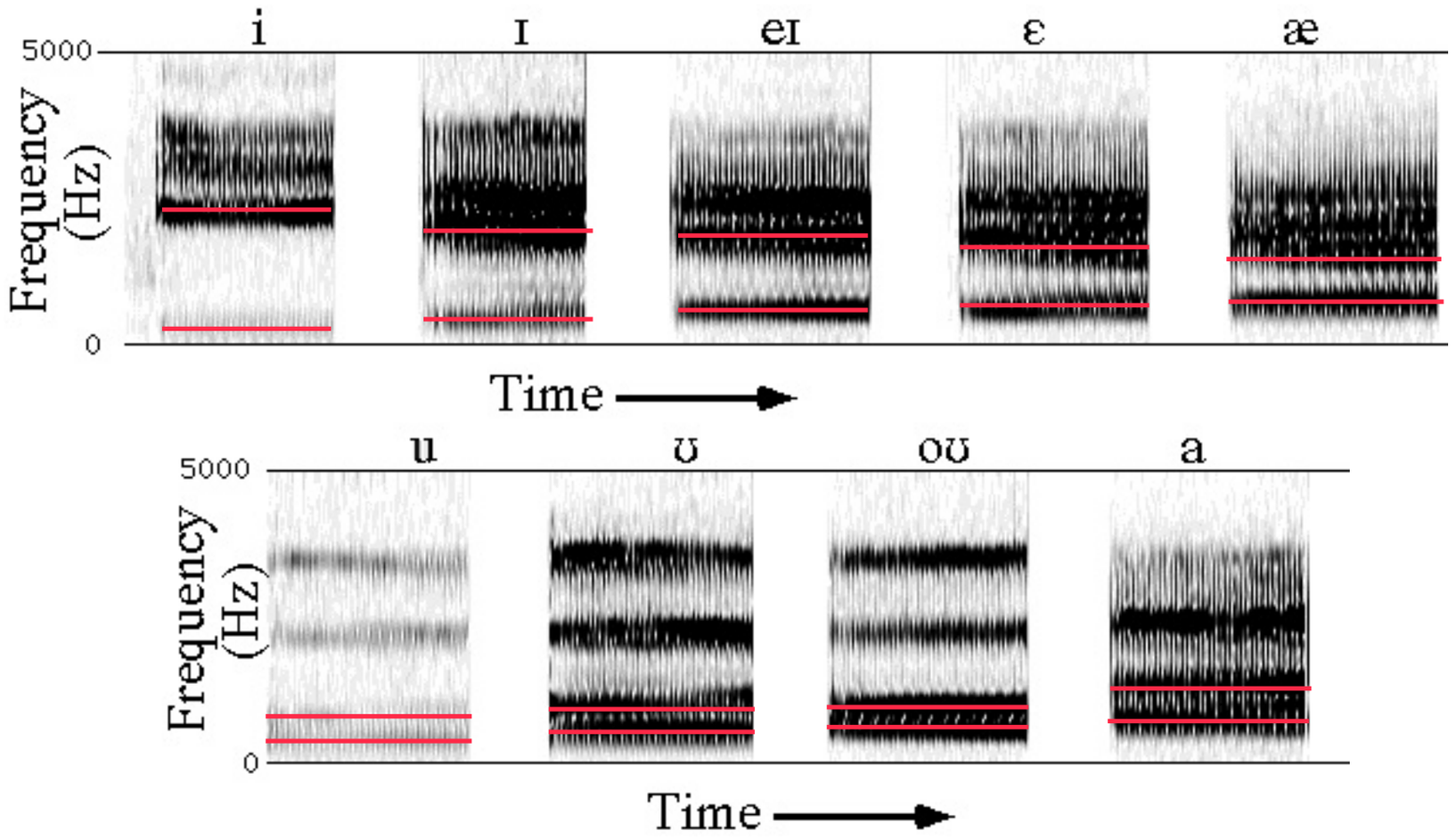
Sustained noisy energy

Voiced vowel

Regular harmonic bands

Simplified, synthetic examples — real speech patterns may vary.

Formants



F1 and F2 are important acoustic cues for distinguishing vowels.

Vowel Space & Formants

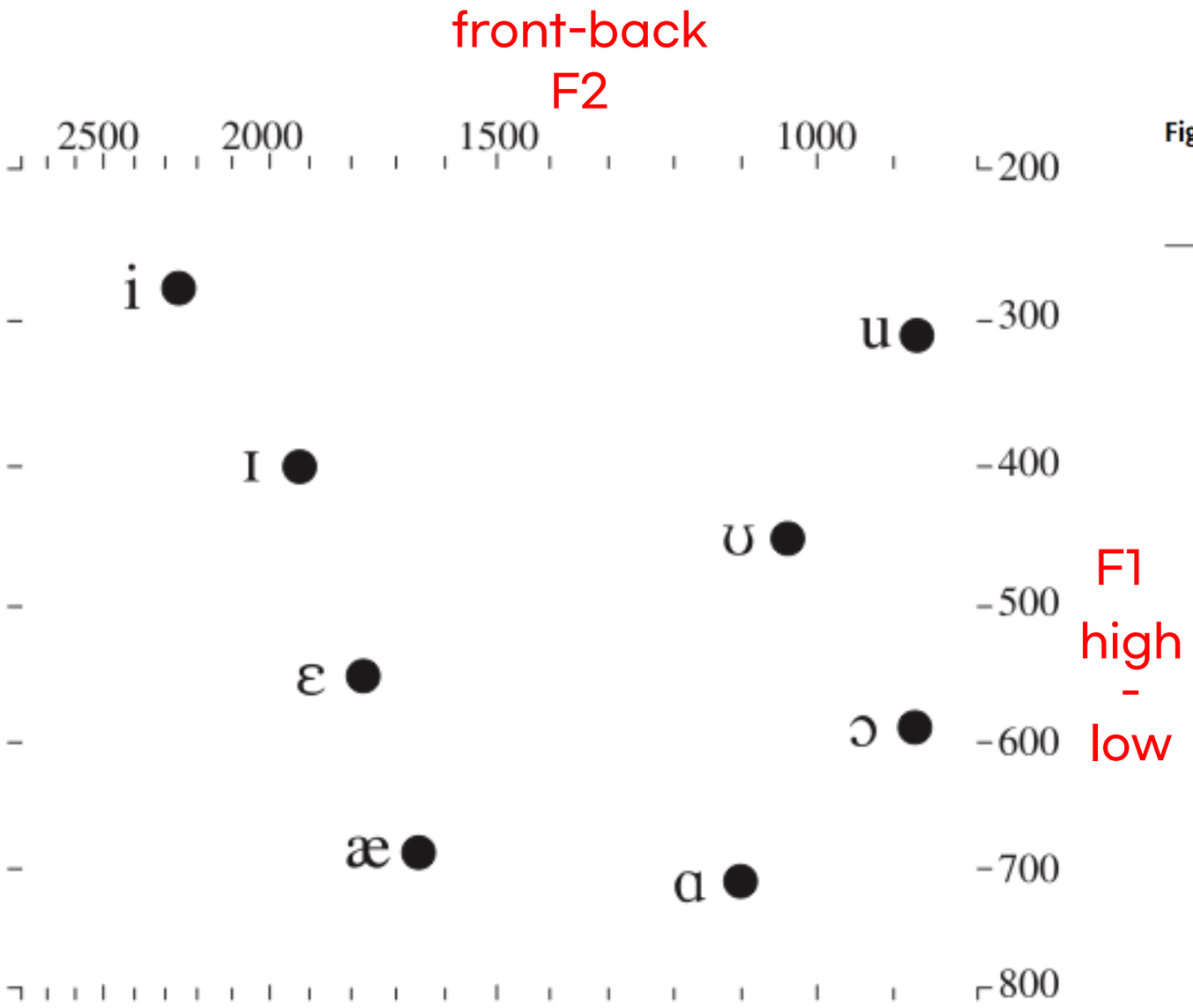


Figure 1.12 The positions of the vocal organs for the vowels in the words 1 *heed*, 2 *hid*, 3 *head*, 4 *had*, 5 *father*, 6 *good*, 7 *food*. The lip positions for vowels 2, 3, and 4 are between those shown for 1 and 5. The lip position for vowel 6 is between those shown for 1 and 7.

